Large amplitude solution of BGK model

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Outline

1. Boltzmann - BGK model
2. Large amplitude solution of BGK model

Boltzmann - BGK model

Boltzmann Equation (Ludwig Boltzmann (1872))

- Boltzmann equation

$$\partial_t F + \underbrace{v \cdot \nabla_x F}_{transport} = \underbrace{Q(F,F)}_{collision},$$

where

$$Q(F,F) = \int_{\mathbb{R}^3}\int_{\mathbb{S}^2} q(\omega, |v-v_*|)(F(v_*')F(v') - F(v_*)F(v))d\omega dv_*,$$

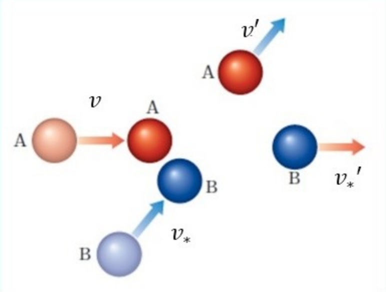


Figure: Elastic collision

Boltzmann BGK model (Bhatnagar-Gross-Krook (1954))

- Relaxation operator $Q(F,F) \to \rho^a(\mathcal{M}(F) - F)$:

$$\partial_t F + v \cdot \nabla_x F = \rho^a(\mathcal{M}(F) - F), \qquad a > 0,$$

where $\mathcal{M}(F)$ is given by

$$\mathcal{M}(F) = \frac{\rho(x,t)}{\sqrt{2\pi T(x,t)}^3} \exp\left(-\frac{|v - U(x,t)|^2}{2T(x,t)}\right).$$

- Density, bulk velocity, temperature:

$$\begin{aligned}
\rho(x,t) &= \int_{\mathbb{R}^3} F(x,v,t)dv, \\
\rho(x,t)U(x,t) &= \int_{\mathbb{R}^3} F(x,v,t)vdv, \\
3\rho(x,t)T(x,t) &= \int_{\mathbb{R}^3} F(x,v,t)|v - U(x,t)|^2 dv.
\end{aligned}$$

Conservation laws and H-theorem

- Conservation law

$$\frac{d}{dt}\int_{\Omega}\int_{\mathbb{R}^3} F\left(1, v, |v|^2\right) dvdx = 0.$$

- H-Theorem

$$\frac{d}{dt}\mathcal{H}(t) = \frac{d}{dt}\int_{\Omega}\int_{\mathbb{R}^3} F \ln F dvdx \leq 0.$$

- Local equilibrium

$$\mathcal{M}(F) = \frac{\rho}{\sqrt{2\pi T}^3} e^{-\frac{|v-U|^2}{2T}}.$$

Property of the Boltzmann and BGK model

- Expand $F = \mu + \sqrt{\mu} f$ around global Maxwellian:

$$\mu(v) = \frac{1}{\sqrt{2\pi}^3} e^{-\frac{|v|^2}{2}}.$$

  - Boltzmann equation near global Maxwellian:

$$\partial_t f + v \cdot \nabla_x f = Lf + \Gamma(f, f).$$

  - BGK model near global Maxwellian:

$$\partial_t f + v \cdot \nabla_x f = Pf - f + \Gamma(f).$$

Existence of the Boltzmann equation

- Renormalized solution [1989 Diperna, Lion. Annals]

- $H^2_{x,v}$ small: [2002,2003 Yan Guo. CPAM, Invent]
- Pointwise behavior: [2004 Tai-Ping Liu, Shih-Hsien Yu. CPAM]
- $L^{\infty,q}_{x,v}$ small: [2010 Yan Guo. ARMA]

- $L^1_x L^\infty_v$ small with small relative entropy [Duan, Huang, Wang, and Yang 2017. ARMA]
- L^2 small L^∞ large [Duan, Wang 2019. Adv Math]

Existence of the BGK model

- Weak sol [1989, Perhtame, JDE]
- Weak sol [1993, Perthame, Pulvirenti. ARMA]

- $H^2_x L^2_v$ small [2010 Yun, JMP]
- L^2 small L^∞ large with additional assumption that (ρ, U, T) is close to $(1,0,1)$. [2017 Duan, Wang, Yang: Sci. Sin. Math.]

Large amplitude solution of BGK model

Main Theorem

Theorem

Let $F_0 = \mu + f_0$ is non-negative, and satisfies

$$\inf_{(t,x)\in[0,\infty)\times\mathbb{T}^3} \int_{\mathbb{R}^3} F_0(x - vt, v)dv \geq C_0,$$

for some positive constant $C_0 > 0$. Then, for any $M_0 > 0$, there exists $\varepsilon = \varepsilon(M_0, C_0)$ such that if initial data f_0 satisfies $(q > 10)$

$$\|f_0\|_{L^{\infty,q}_{x,v}} \leq M_0, \quad \mathcal{E}(F_0) = \int_{\mathbb{T}^3\times\mathbb{R}^3} (F_0 \ln F_0 - \mu \ln \mu)dvdx \leq \varepsilon,$$

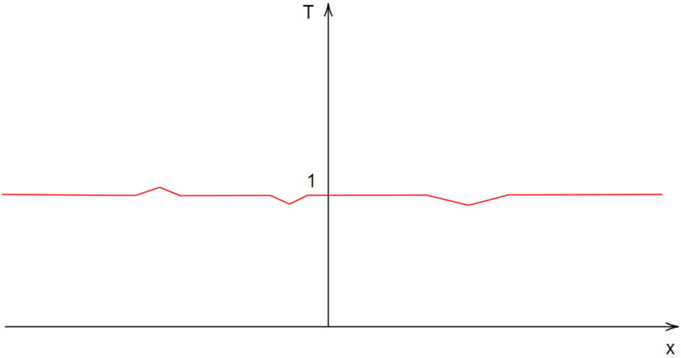
then there exists a unique global-in-time solution to the BGK model. Moreover f satisfies

$$\|f(t)\|_{L^{\infty,q}_{x,v}} \leq C_{M_0} e^{-kt},$$

where C_{M_0} depending on M_0, and k are positive constants.

What is large amplitude?

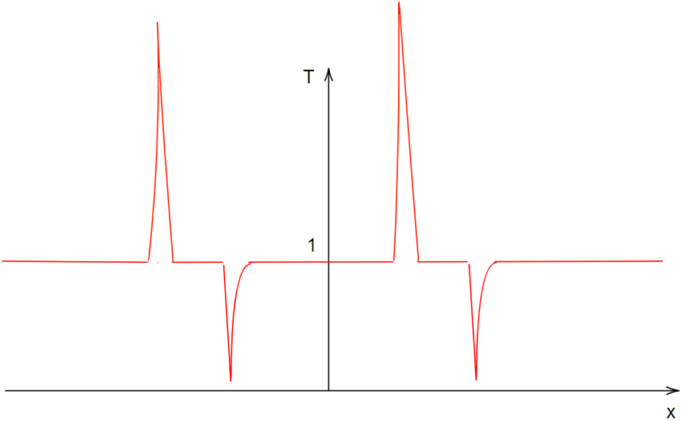
- $H^2_{x,v}$ small or $L^{\infty,q}_{x,v}$ small initial data



What is large amplitude?

- Consider $F_0 = \frac{1}{\sqrt{2\pi T_0}^3} e^{-\frac{|v|^2}{2T_0}}$. Then we have

$$\mathcal{E}(F_0) = \int_{\mathbb{T}^3 \times \mathbb{R}^3} (F_0 \ln F_0 - \mu \ln \mu) dv dx = \int_{\mathbb{T}^3} -\frac{3}{2} \ln T_0 dx \leq \varepsilon,$$



Difficulty

- L^∞ estimate of the Boltzmann equation

$$\|f(t)\|_{L^{\infty,q}_{x,v}} \lesssim C\|f_0\|_{L^{\infty,q}_{x,v}} \left(1+\int_0^t \|f(s)\|_{L^{\infty,q}_{x,v}} ds\right) e^{-\lambda t} + D, \quad D \ll 1.$$

- L^∞ estimate of the BGK model

$$\|f(t)\|_{L^{\infty,q}_{x,v}} \lesssim C(\|f_0\|_{L^{\infty,q}_{x,v}}) \left(1+\int_0^t \left[\|f(s)\|_{L^{\infty,q}_{x,v}}\right]^n ds\right) e^{-\lambda t} + D.$$

Why $F = \mu + f$?

- We define

$$\int_{\mathbb{R}^3} F\left(1, v, \frac{|v|^2-3}{\sqrt{6}}\right) dv = (\rho, \rho U, G).$$

- Global Maxwellian of $(\rho, \rho U, G)$ is equal to $(1, 0, 0)$.
- Make convex combination:

$$\rho_\theta = \theta\rho + (1-\theta), \quad \rho_\theta U_\theta = \theta \rho U, \quad G_\theta = \theta G.$$

Why $F = \mu + f$?

- Nonlinear term of BE:

$$\Gamma(f,f) = \frac{1}{\sqrt{\mu}} Q(\sqrt{\mu}f, \sqrt{\mu}f).$$

- Nonlinear term of BGK:

$$\Gamma(f) = \sum_{1\le i,j\le 5} \int_0^1 \left[\nabla^2_{(\rho_\theta,\rho_\theta U_\theta,G_\theta)}\mathcal{M}(\theta)\right]_{ij} \frac{1}{\sqrt{\mu}}(1-\theta)d\theta \times \int_{\mathbb{R}^3} fe_i\sqrt{\mu}dv \int_{\mathbb{R}^3} fe_j\sqrt{\mu}dv.$$

  where $(e_1,\cdots,e_5) = (1, v, (|v|^2-3)/\sqrt{6})$.

- Unavailable of weighted L^∞ estimate. $\rightarrow$ We use $F = \mu + f$.

Mild form

- Linearization with respect to $F = \mu + f$:

$$\partial_t f + v \cdot \nabla_x f + f = \mathbf{P} f + \Gamma(f),$$

where

$$\mathbf{P} f = \int_{\mathbb{R}^3} f dv \mu + \int_{\mathbb{R}^3} f v dv \cdot (v\mu) + \int_{\mathbb{R}^3} f \frac{|v|^2 - 3}{\sqrt{6}} dv \left(\frac{|v|^2 - 3}{\sqrt{6}} \mu \right),$$

$$\Gamma(f) = \sum_{1 \le i,j \le 5} \int_0^1 \left[\nabla^2_{(\rho_\theta, \rho_\theta U_\theta, G_\theta)} \mathcal{M}(\theta) \right]_{ij} (1-\theta) d\theta \int_{\mathbb{R}^3} f e_i dv \int_{\mathbb{R}^3} f e_j dv.$$

- Mild form:

$$\begin{aligned} f(t,x,v) &= e^{-t} f_0(x - vt, v) \\ &\quad + \int_0^t e^{-(t-s)} (\mathbf{P} f + \Gamma(f))(s, x - v(t-s), v) ds. \end{aligned}$$

Sketch of Proof

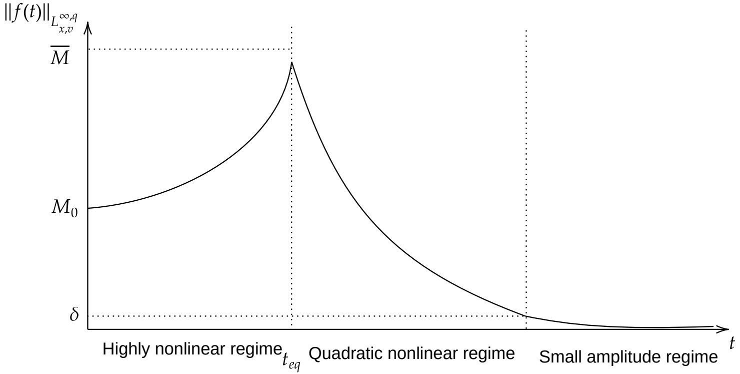


Figure: Three different regimes

Sketch of Proof

1. If $\sup_{0\leq s\leq t_*}\|f(s)\|_{L^{\infty,q}_{x,v}} \leq \overline{M}$, then we have t_{eq} such that $|(\rho,U,T)-(1,0,1)| \leq \delta$ for $t \in [t_{eq},\infty]$. We can reduce the nonlinear term after t_{eq}.
2. Estimate growth of the nonlinear term until t_{eq}.
3. L^∞ decaying estimate. Choosing $\overline{M}$, we can close the argument.
4. Smallness problem

Sketch of Proof (Step 1)

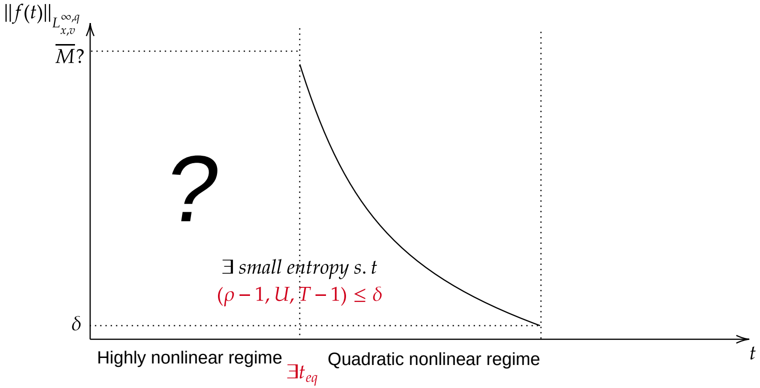


Figure: Step 1

Sketch of Proof of Step 1

Lemma

Under a priori assumption

$$\sup_{0\le s\le t_*} \|f(s)\|_{L^{\infty,q}_{x,v}} \le \overline{M}, \quad \textit{for} \quad q > 10,$$

there exists $\epsilon_{(\overline{M},t_*,\delta)}$ *such that if* $\mathcal{E}(F_0) \le \epsilon_{(\overline{M},t_*,\delta)}$, *then the following estimate holds on* $t \in [0, t_*]$*:*

$$\left| \int_{\mathbb{R}^3} \langle v \rangle^2 f(t,x,v)\,dv \right| \le C_q M_0 e^{-t} + \frac{3}{4}\delta.$$

Sketch of Proof of Step 1

- Divide the velocity region and put the mild form:

$$
\begin{aligned}
\left|\int_{\mathbb{R}^3}\langle v\rangle^2 f(t,x,v)dv\right| &\leq \frac{C_q}{N^{q-5}}\overline{M} + e^{-t}\int_{|v|\leq N}\langle v\rangle^2|f_0(x-vt,v)|dv \\
&+ \int_0^t e^{-(t-s)}\int_{|v|\leq N}\langle v\rangle^2|\mathbf{P}f(s,x-v(t-s),v)|dvds \\
&+ \int_0^t e^{-(t-s)}\int_{|v|\leq N}\langle v\rangle^2|\Gamma(f)(s,x-v(t-s),v)|dvds.
\end{aligned}
$$

for $t\in[0,t_*]$ and arbitrary $N>0$.

Sketch of Proof of Step 1

We consider $\mathbf{P}f$ part:

$$\int_0^t \int_{|v|\leq N} e^{-(t-s)} \langle v \rangle^2 |\mathbf{P}f(s, x - v(t-s), v)| dvds$$

Divide the integral region

$$\begin{aligned} I &= \int_{t-\lambda}^t \int_{|v|\leq N} \int_{\mathbb{R}^3} \cdots dudvds, \\ II &= \int_0^{t-\lambda} \int_{|v|\leq N} \int_{|u|\geq 2N} \cdots dvds, \\ III &= \int_0^{t-\lambda} \int_{|v|\leq N} \int_{|u|\leq 2N} \cdots dudvds. \end{aligned}$$

Sketch of Proof of Step 1

Estimate for III: Using $\langle v\rangle^2 \leq N^2$ and $(1+|u|+|u|^2) \leq 4N^2$, we have

$$III \leq 4N^6 \int_0^{t-\lambda} e^{-(t-s)} \int_{|v|\leq N} \int_{|u|\leq 2N} |f(s, x - v(t-s), u)| du dv ds.$$

Change of variable $y = x - v(t-s)$ with $dy = -(t-s)^3 dv$:

$$III \leq 4N^6 \int_0^{t-\lambda} e^{-(t-s)} \frac{1+(N(t-s))^3}{(t-s)^3} \int_{\mathbb{T}^3} \int_{|u|\leq 2N} |f(s, y, u)| du dy ds.$$

Sketch of Proof of Step 1

Divide the integral into $\mathbf{1}_{|f(s,y,u)|>\mu(u)}$ and $\mathbf{1}_{|f(s,y,u)|\leq\mu(u)}$:

$$
\begin{aligned}
III \leq 4N^6 \int_0^{t-\lambda} e^{-(t-s)} \frac{1+(N(t-s))^3}{(t-s)^3} \\
\times \left(\int_{\mathbb{T}^3} \int_{|u|\leq 2N} |f(s,y,u)| \mathbf{1}_{|f(s,y,u)|>\mu(u)} dudy \right. \\
\left. + \int_{\mathbb{T}^3} \int_{|u|\leq 2N} \frac{1}{\sqrt{\mu(u)}} |f(s,y,u)| \mathbf{1}_{|f(s,y,u)|\leq\mu(u)} dudy \right) ds.
\end{aligned}
$$

Note the relation of entropy:

$$
\begin{aligned}
\mathcal{E}(F_0) \geq \int_{\mathbb{T}^3\times\mathbb{R}^3} \frac{1}{4} |f(t,x,v)| \mathbf{1}_{|f(t,x,v)|>\mu(v)} dvdx \\
+ \int_{\mathbb{T}^3\times\mathbb{R}^3} \frac{1}{4\mu(v)} |f(t,x,v)|^2 \mathbf{1}_{|f(t,x,v)|\leq\mu(v)} dvdx.
\end{aligned}
$$

Sketch of Proof of Step 1

Thus we have

$$III \le CN^6(\lambda^{-2} + N^3)\left(\mathcal{E}(F_0) + N^{\frac{3}{2}}\sqrt{\mathcal{E}(F_0)}\right).$$

Result:

$$\begin{aligned} I + II + III \le\ & C_q N^5(1 - e^{-\lambda})\overline{M} + \frac{C_q}{N^{q-10}}\overline{M} \\ & + CN^6(\lambda^{-2} + N^3)\left(\mathcal{E}(F_0) + N^{\frac{3}{2}}\sqrt{\mathcal{E}(F_0)}\right). \end{aligned}$$

Sketch of Proof (Step 2)

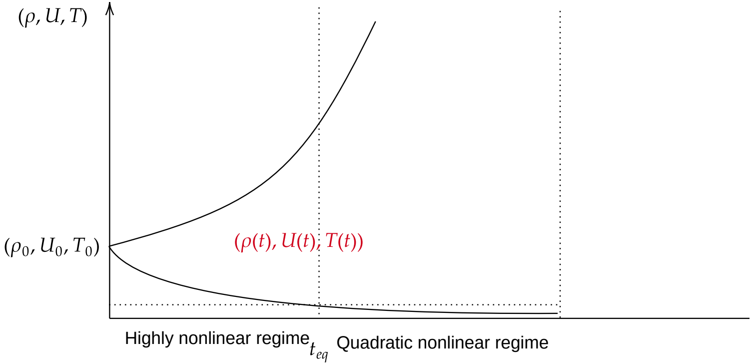


Figure: Step 2

Sketch of Proof (Step 2)

(Step 2) Estimate growth until t_{eq}.
Upper bound of the macroscopic fields:

$$
\begin{aligned}
&(1)\ C_0 e^{-C_q M_0^a t} \leq \rho(t,x) \leq C_q M_0,\\
&(2)\ |U(t,x)| \leq C_q M_0 e^{C_q M_0^a t},\\
&(3)\ C M_0^{-\frac{2}{3}} e^{-\frac{4}{3} C_q M_0^a t} \leq T(t,x) \leq C_q M_0 e^{C_q M_0^a t}.
\end{aligned}
$$

We can reduce the nonlinear term to quadratic nonlinearity.
Estimate of the nonlinear term:

$$
\|\Gamma(f)(t,x,v)\|_{L^{\infty,q}_{x,v}} \leq C_q M_0^n e^{C_q M_0^a t} \|f(t)\|_{L^{\infty,q}_{x,v}} \int_{\mathbb{R}^3} |f(u)|(1+|u|^2)du.
$$

Sketch of Proof (Step 3)

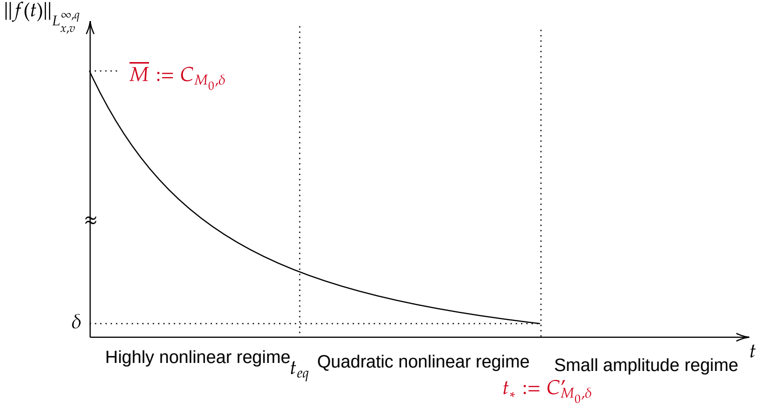


Figure: Step 3

Sketch of Proof (Step 3)

(Step 3) L^∞ estimate

$$\|f(t)\|_{L^{\infty,q}_{x,v}} \leq C_q e^{C_q M_0^a t_{eq}} M_0^{n+1} e^{-t/2}\left(1+\int_0^t \|f(s)\|_{L^{\infty,q}_{x,v}} ds\right) + D.$$

We have

$$\|f(t)\|_{L^{\infty,q}_{x,v}} \leq C_q e^{C_q M_0^a t_{eq}} M_0^{n+1} \exp\left\{2C_q e^{C_q M_0^a t_{eq}} M_0^{n+1}\right\}(1+Dt)e^{-t/2} + D.$$

Choose $\overline{M}$:

$$\overline{M} := 4C_q M_0^{n+1} \exp\left\{C_q M_0^a t_{eq} + 2C_q e^{C_q M_0^a t_{eq}} M_0^{n+1}\right\}.$$

We can extend the local in time unique solution until t_*.

Sketch of Proof (Step 4)

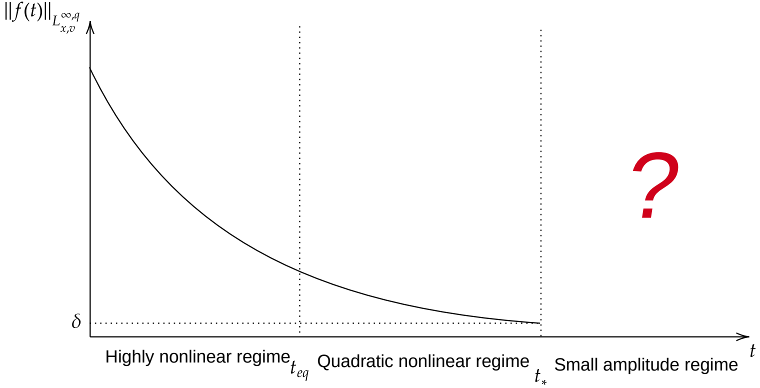


Figure: Step 4

Sketch of Proof (Step 4)

Lemma

There exists $\delta > 0$ such that if $\|f_0\|_{L^{\infty,q}_{x,v}} \leq \delta$, then there exists a unique global solution of the BGK model. Moreover, the following holds:

1. *The solution is non-negative: $F(t,x,v) = \mu(v) + f(t,x,v) \geq 0$.*
2. *The perturbation decays exponentially:*

$$\|f(t)\|_{L^{\infty,q}_{x,v}} \leq C_q \delta e^{-kt}.$$

for positive constants k and C_q.

Split the system:

$$\begin{aligned} &\partial_t f_1 + v \cdot \nabla_x f_1 + f_1 = \Gamma(f_1 + f_2), \qquad && f_1(0,x,v) = f_0(x,v), \\ &\partial_t f_2 + v \cdot \nabla_x f_2 = (\mathbf{P} f_2 - f_2) + \mathbf{P} f_1, \qquad && f_2(0,x,v) = 0, \end{aligned}$$

Summary and future works

- We found t_{eq} such that $|(\rho, U, T) - (1,0,1)| \leq \delta$ for $t \in [t_{eq}, \infty]$ under a priori assumption.
- We can measure the amount of increase until the finite time t_{eq}.

Thank you for attention!